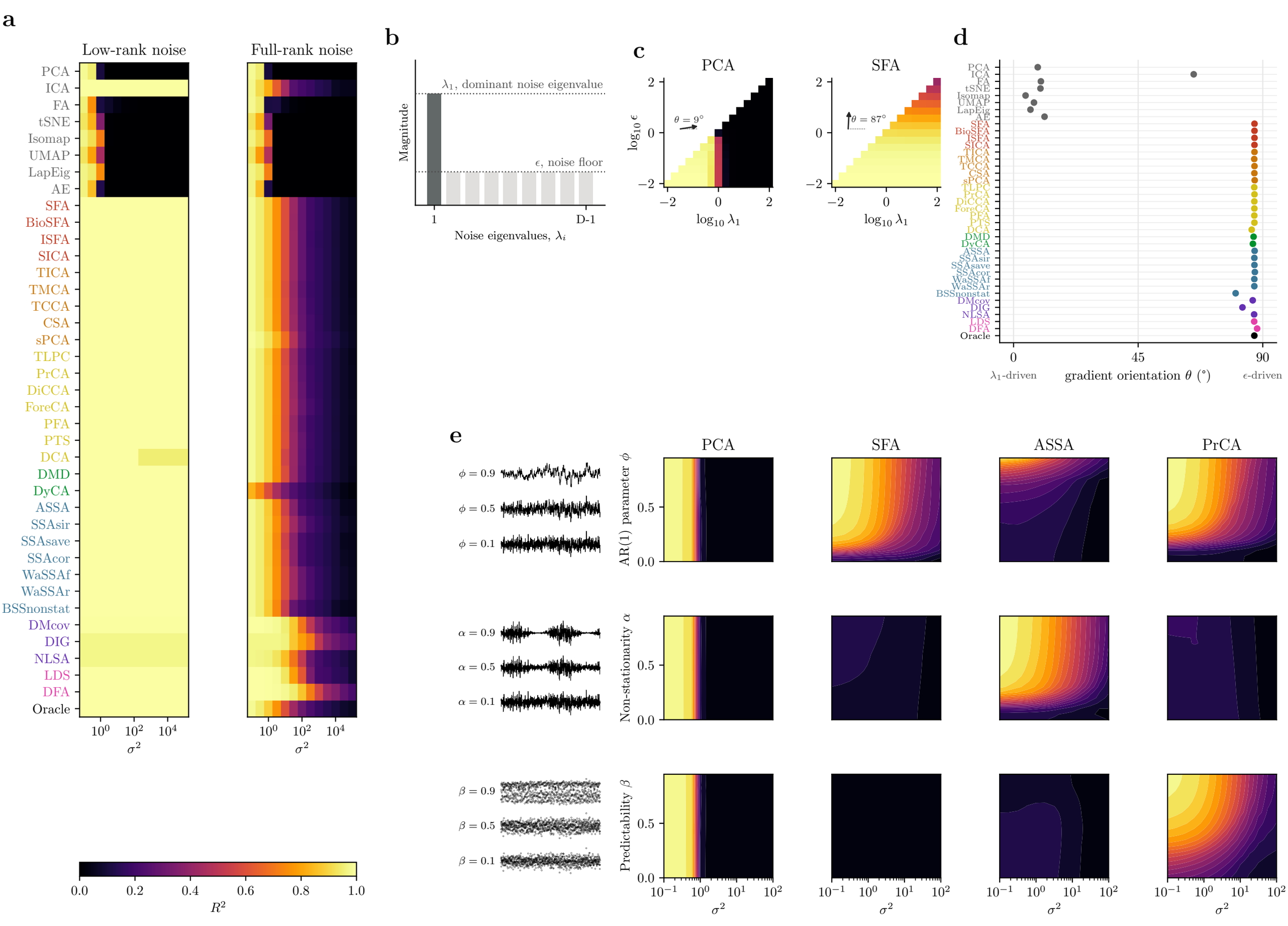

a
Low-rank noise
Full-rank noise
PCA
ICA
FA
tSNE
Isomap
UMAP
LapEig
AE
SFA
BioSFA
ISFA
SICA
TICA
TMCA
TCCA
CSA
sPCA
TLPC
PrCA
DiCCA
ForeCA
PFA
PTS
DCA
DMD
DyCA
ASSA
SSAsir
SSAsave
SSAcor
WaSSAf
WaSSAr
BSSnonstat
DMcov
DIG
NLSA
LDS
DFA
Oracle
10^0 10^2 10^4
σ^2
0.0 0.2 0.4 0.6 0.8 1.0
R^2
b
λ_1, dominant noise eigenvalue
ϵ, noise floor
Magnitude
1
D-1
Noise eigenvalues, λ_i
c
PCA
SFA
$\theta = 9°$
$\theta = 87°$
$\log_{10} \epsilon$
$\log_{10} \lambda_1$
2
0
−2
d
0
45
90
λ_1-driven
gradient orientation θ (°)
ϵ-driven
e
$\phi = 0.9$
$\phi = 0.5$
$\phi = 0.1$
$\alpha = 0.9$
$\alpha = 0.5$
$\alpha = 0.1$
$\beta = 0.9$
$\beta = 0.5$
$\beta = 0.1$
PCA
SFA
ASSA
PrCA
AR(1) parameter ϕ
Non-stationarity α
Predictability β
0.0
0.5
10^{-1} 10^0 10^1 10^2
σ^2